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Prescription for Change: Illuminating the Shadows of the Pharmaceutical Insulin Industry

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Insulin, the drug that is used to regulate your insulin levels was created in the early 1920s by two Canadian scientists, with the purpose of creating an affordable and assessable treatment for all those who have diabetes. The patent was famous and sold for \$1 to the University of Toronto a direct quote from one of the creators was “Insulin doesn’t belong to me; it belongs to the world.(Diabetes UK n.d.)” The intention behind the invention was lost in translation in the American markets,

as profit maximization and oligopolization of the drug has led the opposite of what the inventors of insulin sought to do.

Rising Diabetes Rates

Demand can only exist if there exists a need for the drug. To understand the exploitation, we must first understand what led to the cause of the drug becoming exploited. Since 1980, both diabetes and obesity rates have been increasing at a high rate, and most experts believe that there is a correlation between the two. Some driving factors that have caused this are lifestyle changes that promote inactivity, ingestion of excess calories, ageing populations, and increased promotion of unhealthy foods. This consistent and constant rise has become an epidemic, or well epidemic that has directly caused the other (Zimmet 2017). Within the United States, diabetes rates have increased from around 2.7% of the total population in 1980 to a staggering 11.3% of the total population in 2022, with around 2.5% of cases being undiagnosed (CDC 2022). What these numbers represent is a staggering 11% or close to 37 million Americans are dependent on a daily dose to regulate their insulin levels as for them the drug is a necessity to live. If the rates don't seem too bad now, projections indicate that the number of diabetics will skyrocket to around 54.9 million, which is a 67% increase in total cases (Rowley 2017). These numbers indicate that at least 40 million people are dependent on this drug to survive and make it to the next day, which is precisely what is being exploited.

Market Dominance Within the Sector

The term oligopoly means that there is a small concentration of firms that have a high market share within the market and limited choices for consumers. This type of market structure often results in firms closely monitoring and responding to each other's actions, creating an environment where they may engage in price-fixing, collusion, or other anti-competitive practices to maintain their dominant positions and maximize profits. As a result, consumers may find themselves with

fewer alternatives and paying higher prices for the goods or services offered by these powerful oligopolistic firms. An example that I'm sure hits close to home would be the telecom markets that are highly dominated by Bell, Rogers, and Telus. These three firms dominate the market and cause Canadian phone prices to be some of the highest in the world and they have created such high barriers to entry for newer firms trying to enter the market. In the American insulin market, Eli Lilly, Novo Nordisk, and Sanofi collectively control a significant 90% share, demonstrating their dominant presence in the industry.

Effects Of the Oligopoly

Price Gauging

When a small coalition of companies owns a vast majority of the market share, this directly leads to a bidding war between the group, which increases profits for the three firms while hurting the consumer. In ordinary circumstances, when the price increases too much this leads to the demand for the product decreasing but in the case of medicine, demand will stay the same. These companies are aware of this, and their unique position within the market. As a result of this, American insulin prices are the highest in the world by a large margin. Over the past 20 years, the price of American insulin has increased at a monumental rate, not only in comparison to the rest of the world but in relation to American prices as well. The cost of a vial of insulin was \$21 in 1996, the cost for that same vial in 2023 would be upwards of \$250 per vial. If you were to think that the price of production was to increase along with the price, meaning the increase in price is due to an increase in production then you would be surely mistaken. The cost of production for a vial is between \$2-\$4, meaning the only thing that has increased is the ever-increasing bottom lines of the three firms that have oligopolies in this sector of pharmaceuticals. On average, the cost of American insulin is seven to eight times more expensive than the rest of the world (Yale School of Medicine 2023).

Patent Evergreening

In most markets, whenever there is a high price, a firm always comes in to undercut and to capitalize on the high prices, giving consumers an incentive to use their products over the firm with higher prices. To get this point for the competitors of Eli Lilly, Novo Nordisk, and Sanofi the competing firms must be able to introduce their product into the market in the first place. These firms use a strategy called patent evergreening, which essentially means drug manufacturers repeatedly make minor changes to the patent formula, which means that they must apply for new patents each time a change. This not only slows down the process for competing firms to bring their own products into the market, but it actively discourages as the big three firms actively pay off their competitors to not copy their drugs for a specific time (Team 2021). As these firms have created all these hurdles, this creates a lack of competition, leaving no incentive for other firms to come in to “undercut” as the cost of research, regulatory affairs, and approval process is a lengthy and expensive process with no guarantee of a piece of the pie.

Patient Consequences

Within oligopolistic models, consumers are often the ones who bear the brunt of exploitative market practices. While high prices and limited options in the Canadian telecom market may be frustrating, they are far from life-or-death situations. In the realm of pharmaceuticals, and particularly for medications like insulin, the consequences of restricted access can indeed be the difference between life and death. The lack of effective regulation and control over drug pricing in the United States has given rise to a dire situation known as insulin rationing, where individuals are forced to make a vial of insulin last longer than it should, often with catastrophic consequences.

Many individuals with diabetes in America find themselves making impossible choices. They are faced with the decision to allocate their limited financial resources between essentials like food, housing, and insulin. When they cannot afford their insulin, patients may resort to using less than the prescribed dosage, skipping doses, or even discontinuing their treatment altogether. The consequences of such choices are dire. This self-rationing can lead to unstable blood sugar levels, increasing the risk of life-threatening complications and hospitalization. Insulin rationing disproportionately affects marginalized and vulnerable populations. Low-income individuals, those without adequate health insurance, and underprivileged communities are hit the hardest. Inequitable access to essential medications like insulin exacerbates existing healthcare disparities, leading to poorer health outcomes and increased healthcare costs in the long run. The repercussions of insulin rationing extend beyond immediate health concerns. When individuals are unable to afford or adequately access insulin, they often experience a cascade of negative effects. This includes missed workdays due to illness, a decreased quality of life, increased emergency room visits, and a greater strain on the healthcare system. The financial stress and emotional burden of not being able to afford a life-saving medication can have severe psychological consequences. Patients may experience anxiety, depression, and a sense of hopelessness, further compromising their overall well-being.

Conclusion

The story of insulin in America, from its altruistic beginnings to its contemporary commercial exploitation, exposes the grim realities of a healthcare system where patients are caught in the crossfire of profit-driven pharmaceutical practices. The soaring demand for insulin due to rising diabetes rates will cause the big three firms to have a greater incentive to continue with their oligopolistic actions, if actions are not taken then people will continue to suffer not only physically but financially. This showcases a pressing need for regulatory reforms to

ensure accessible and affordable insulin for all is evident, reminding us that the primary purpose of medicine should be to save lives rather than maximize profits.

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